hw4_svd_transformation

March 28, 2026

1 Readme

Places where solutions are required are marked with **#TODO**
You will **NOT** need to modify any section not marked as #TODO to answer this question.
Make sure the helper file. svd_transformation_helper.py is in the same folder as this .ipynb
Make sure you have numpy, matplotlib and itertools packages installed for python

1.0.1 In this notebook:

Part (b) has 3 subparts i, ii, and iii
Part (c) has 4 subparts i, ii, iii and iv
Part (d) has 2 subparts i,ii
Part (e) has only 1 subpart

[1]:
```
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
!gdown 1cV2RxKzE-02nGVMXi-O92dytf6-Cg-3D -O svd_transformation_helper.py
from svd_transformation_helper import visualize_function
from svd_transformation_helper import matrix_equals, is_orthonormal
```

```
Downloading…
From (original):
https://drive.google.com/uc?id=1cV2RxKzE-02nGVMXi-O92dytf6-Cg-3D
From (redirected): https://drive.google.com/uc?id=1cV2RxKzE-02nGVMXi-O92dytf6-
Cg-3D&confirm=t&uuid=42ee7f56-6ab3-4559-adb0-9ea8082bad12
To: /content/svd_transformation_helper.py
   0% 0.00/4.63k [00:00<?, ?B/s] 100% 4.63k/4.63k [00:00<00:00, 10.2MB/s]
```

[2]:
```
DISABLE_CHECKS = False #Set this to True only if you get Value Errors about␣
↪inputs even
#when you are sure that what you are inputting is correct.
#WARNING: Setting this to True and entering wrong inputs can lead to all kinds␣
↪of crazy results/errors


def visualize(U = np.identity(2), D = np.ones(2), VT = np.identity(2),␣
↪num_grid_points_per_dim = 200,\
```

```
    disable_checks = DISABLE_CHECKS, show_original = True, show_VT = True,␣
↪show_DVT = True, show_UDVT = True):
    '''
    Inputs:
    A has singular value decomposition A = U np.diag(D) VT
    U: 2 x 2 orthonormal matrix represented as a np.array of shape (2,2)
    D: Diagonal entries corresponding to the diagonal matrix in SVD represented␣
↪as a np.array of shape (2,)
    VT: 2 x 2 orthonormal matrix represented as a np.array of shape (2,2)
    num_grid_points_per_dim: Spacing of points used to represent circle␣
↪(Decrease this if plotting is slow)
    disable_checks: If False then have checks in  place to make sure dimensions␣
↪of VT, U are correct, etc.
    show_original: If True plots original unit circle and basis vectors
    show_VT: If True plots transformation by VT
    show_DVT: If True plots transformation by DVT
    show_UDVT: If True plots transformation by UDVT
    '''

    visualize_function(U=U, D=D, VT=VT,␣
↪num_grid_points_per_dim=num_grid_points_per_dim,␣
↪disable_checks=disable_checks,\
                       show_original=show_original, show_VT=show_VT,␣
↪show_DVT=show_DVT, show_UDVT=show_UDVT)
```

2 Effect of the linear transformation by an orthonormal matrix V^T

A 2 x 2 orthonormal matrix can be viewed as a linear transformation that performs some combination of rotations and reflections. Note that both rotation and reflection are operations that preserve the length of vectors and the angle between them.

2.1 V^T as a rotation matrix

First we set V^T as a counter-clockwise rotation matrix.

2.1.1 (b) i: Fill in the function "get_RCC(theta)" to return a 2 x 2 matrix that, when applied to a vector x, rotates it by theta radians counter clockwise.

Example: If $V^T = RCC\left(\frac{\pi}{4}\right)$ and x $= \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, then,

$$V^T \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

[9]:
```
def get_RCC(theta):
    '''
    Returns a 2 x 2 orthonormal matrix that rotates x by theta radians␣
↪counter-clockwise
    '''

    # RCC = np.identity(2) #TODO: Solution to (b) i. Change this line by␣
↪filling in the correct reflection matrix
    RCC = np.array([[np.cos(theta), np.sin(-theta)], [np.sin(theta), np.
↪cos(theta)]])
    ␣
↪##########################################################################################
    #Some assertions (WARNING: Do not modify below code)
    if DISABLE_CHECKS is False:
        if not isinstance(RCC, np.ndarray) or isinstance(RCC, np.matrix):
                raise ValueError('RCC must be a np.ndarray')
        if len(RCC.shape) != 2 or (RCC.shape != np.array([2,2])).any():
                raise ValueError('RCC must have shape [2,2]')
    return RCC
```

2.1.2 get_RCC(theta) function test

If the function get_RCC(theta) is defined correctly then you should not get any ERROR statement here.

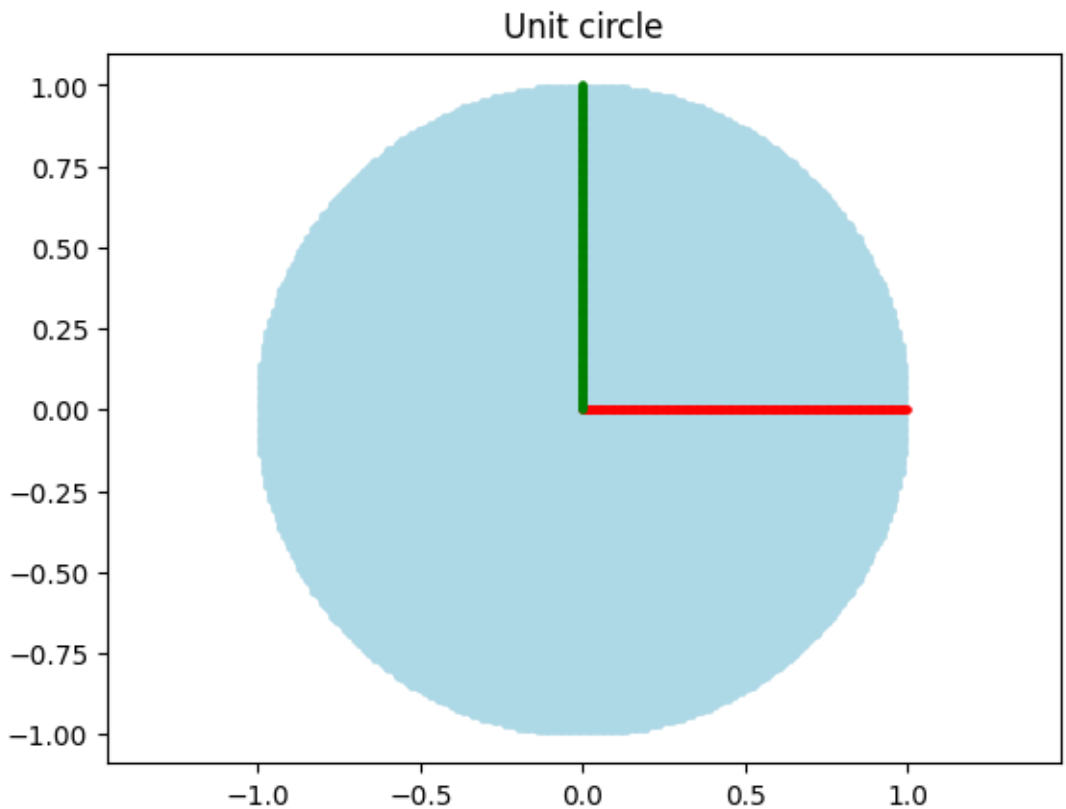
[10]:
```
x = np.array([[1,0]]).T
V_test = get_RCC(np.pi/4)
y = V_test @ x
expected_y = np.array([[1/np.sqrt(2), 1/np.sqrt(2)]]).T
print("y:")
print(y)
print("Expected y:")
print(expected_y)
if not matrix_equals(y, expected_y):
    print("ERROR: y does not match expected_y. Check if function get_RCC(theta)␣
↪is completed correctly")
else:
    print("MATCHED: y matches expected_y!")
```

```
y:
[[0.70710678]
 [0.70710678]]
Expected y:
[[0.70710678]
 [0.70710678]]
MATCHED: y matches expected_y!
```

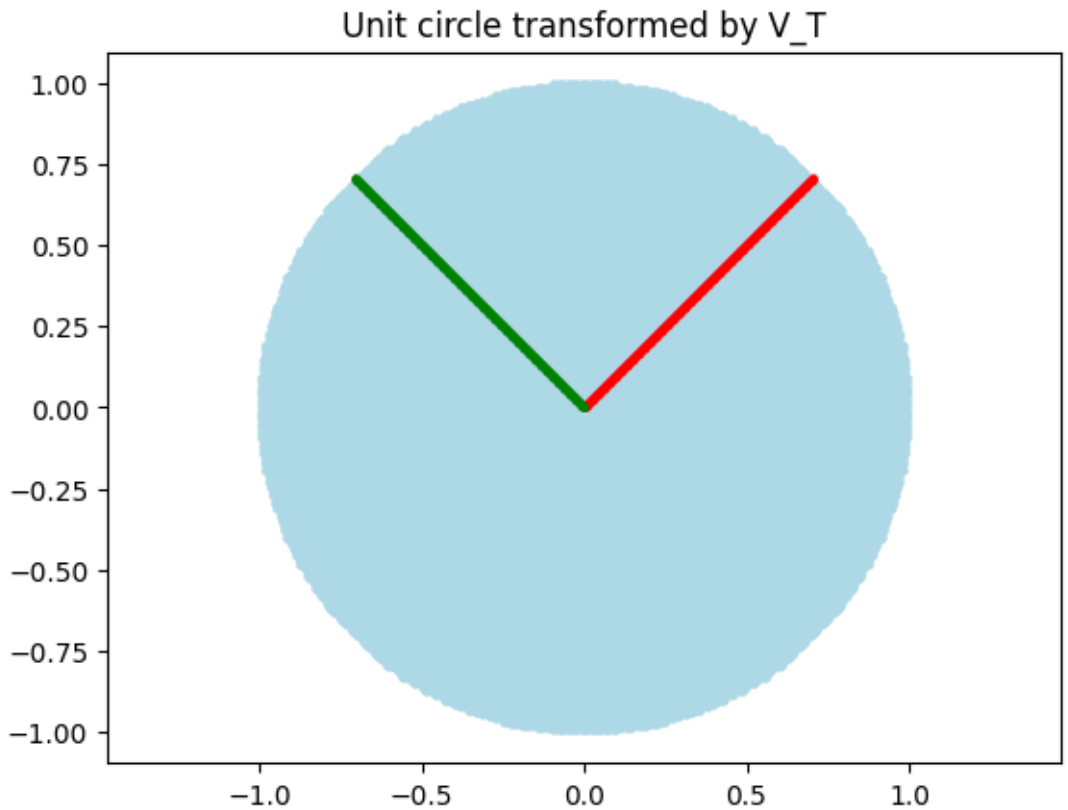
Next we observe how V^T transforms the unit circle and unit basis vectors when:

1) $V^T = RCC\left(\frac{\pi}{4}\right)$

[11]:
```
VT_1 = get_RCC(np.pi/4)
visualize(VT = VT_1, show_DVT=False, show_UDVT=False)
```

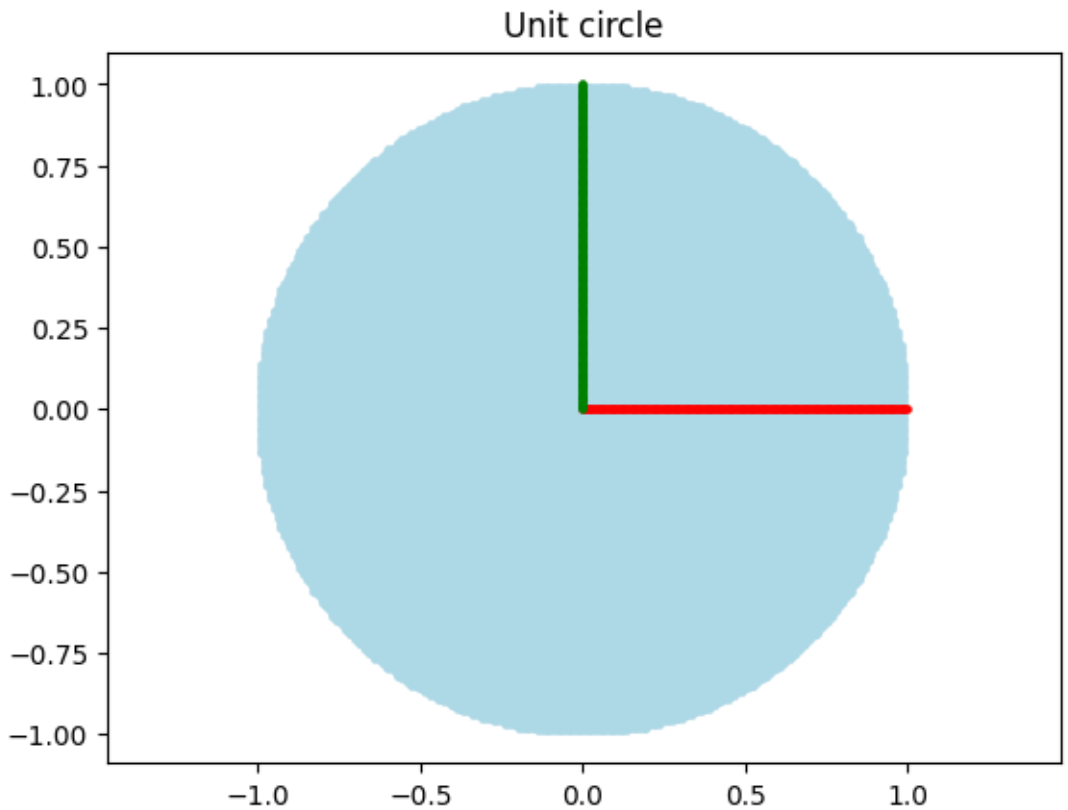



```
WARNING:matplotlib.axes._base:Ignoring fixed x limits to fulfill fixed data
aspect with adjustable data limits.
```

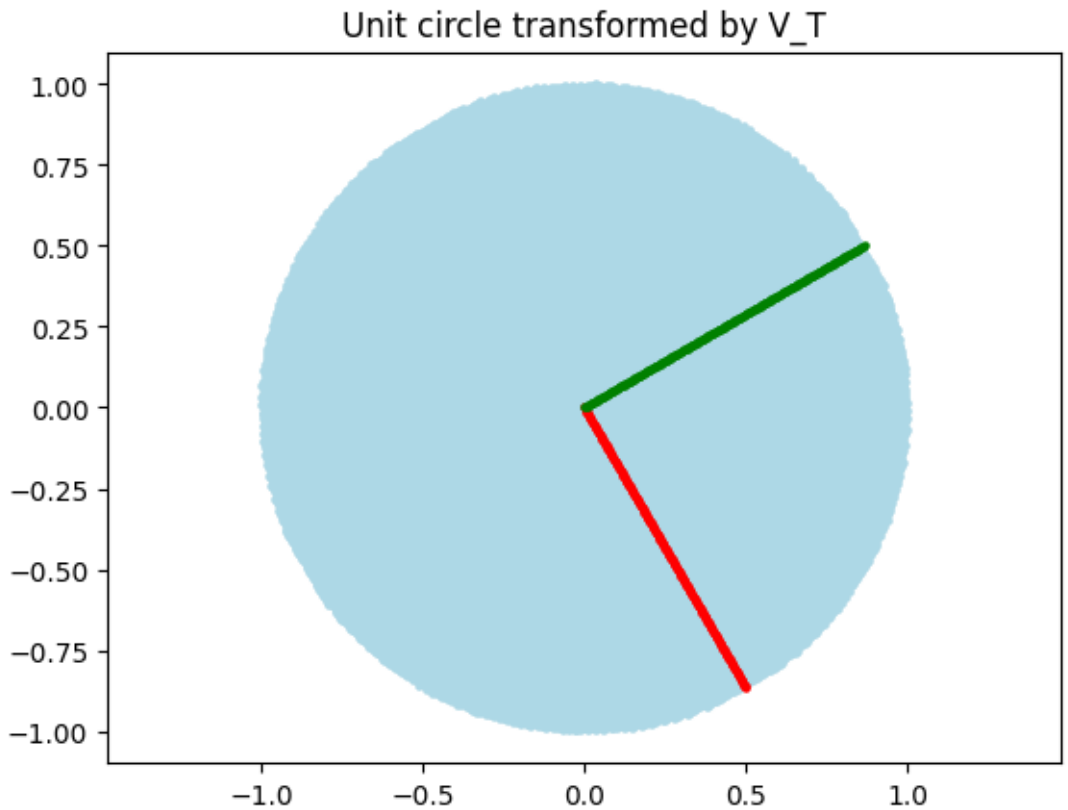


2) $V^T = RCC\left(\frac{-\pi}{3}\right)$

[12]:
```
VT_2 = get_RCC(-np.pi/3)
visualize(VT = VT_2, show_DVT=False, show_UDVT=False)
```




```
WARNING:matplotlib.axes._base:Ignoring fixed x limits to fulfill fixed data
aspect with adjustable data limits.
```



Next we consider the case where V^T is a reflection matrix.

2.2 V^T as a reflection matrix

A reflection matrix is another type of orthonormal matrix.

2.2.1 (b) ii: Fill in the function "get_RFx()" to return a 2 x 2 matrix that when applied to a vector x reflects it about the x-axis.

Example: If $V^T = RFx()$ and $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, then,

$V^T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

[23]:
```
def get_RFx():
    '''
    Returns a 2 x 2 orthonormal matrix that reflects about x-axis
    '''
    # RFx = np.identity(2) #TODO: Solution to (b) ii: change this line by␣
↪filling in the correct reflection matrix
    RFx = np.array([[1, 0],[0, -1]])
```

```
␣
↪##########################################################################################
    #Some assertions (WARNING: Do not modify below code)
    if DISABLE_CHECKS is False:
        if not isinstance(RFx, np.ndarray) or isinstance(RFx, np.matrix):
                raise ValueError('RFx must be a np.ndarray')
        if len(RFx.shape) != 2 or (RFx.shape != np.array([2,2])).any():
                raise ValueError('RFx must have shape [2,2]')
    return RFx
```

2.2.2 get_RFx() function test

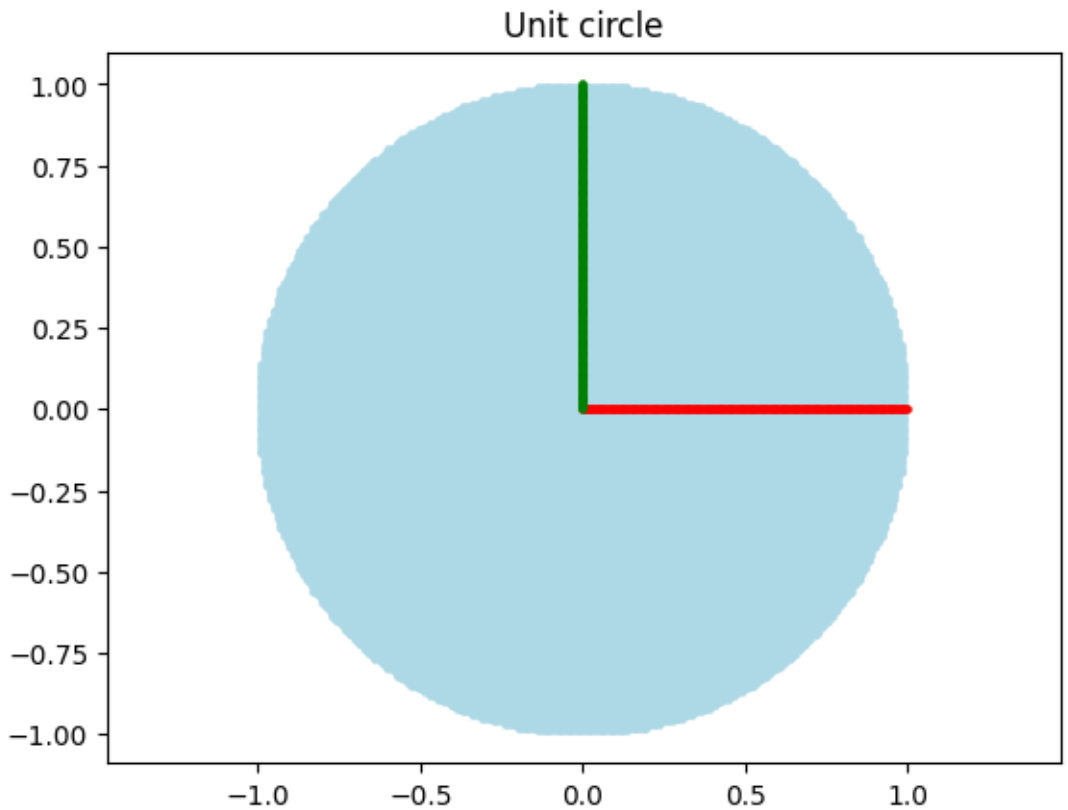
If the function get_RFx() is defined correctly then you should see a MATCHED statement here.

[24]:
```
x = np.array([[1,1]]).T
V_test = get_RFx()
y = V_test @ x
expected_y = np.array([[1, -1]]).T
print("y:")
print(y)
print("Expected y:")
print(expected_y)
if not matrix_equals(y, expected_y):
    print("ERROR: y does not match expected_y. Check if function get_RFx() is␣
↪completed correctly")
else:
    print("MATCHED: y matches expected_y!")
```

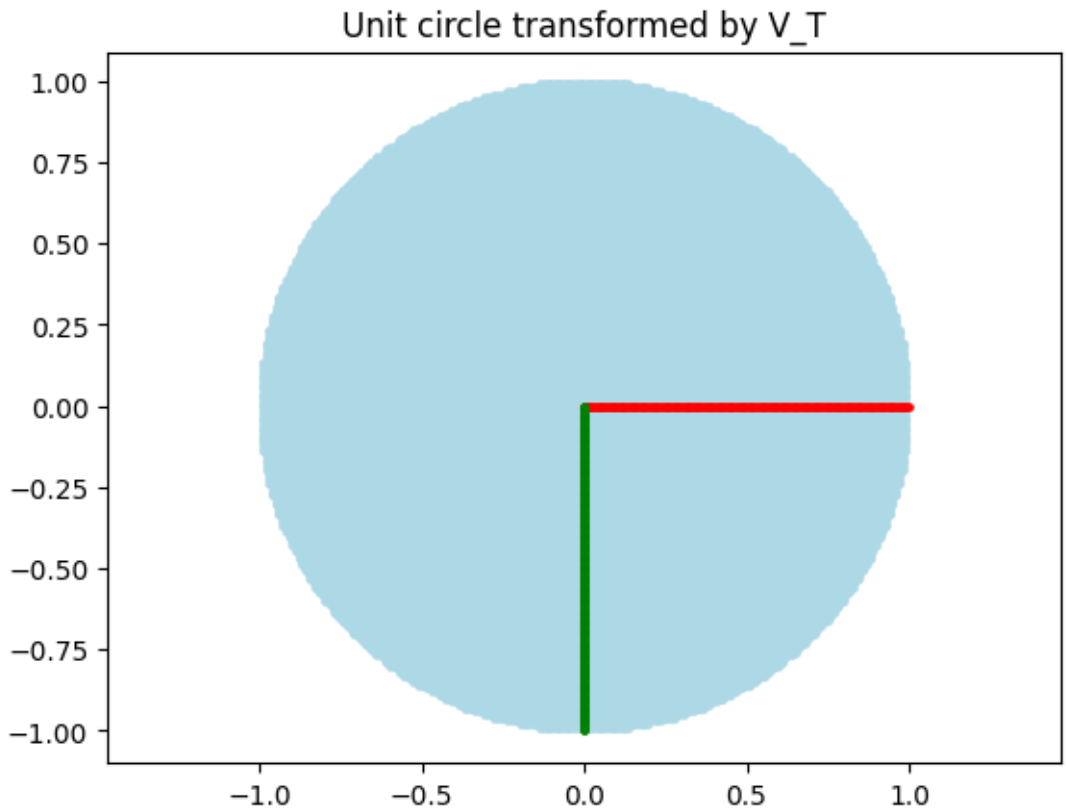
```
y:
[[ 1]
 [-1]]
Expected y:
[[ 1]
 [-1]]
MATCHED: y matches expected_y!
```

$V^T = RFx()$

[25]:
```
VT_3 = get_RFx()
visualize(VT = VT_3, show_DVT=False, show_UDVT=False)
```




```
WARNING:matplotlib.axes._base:Ignoring fixed x limits to fulfill fixed data
aspect with adjustable data limits.
```



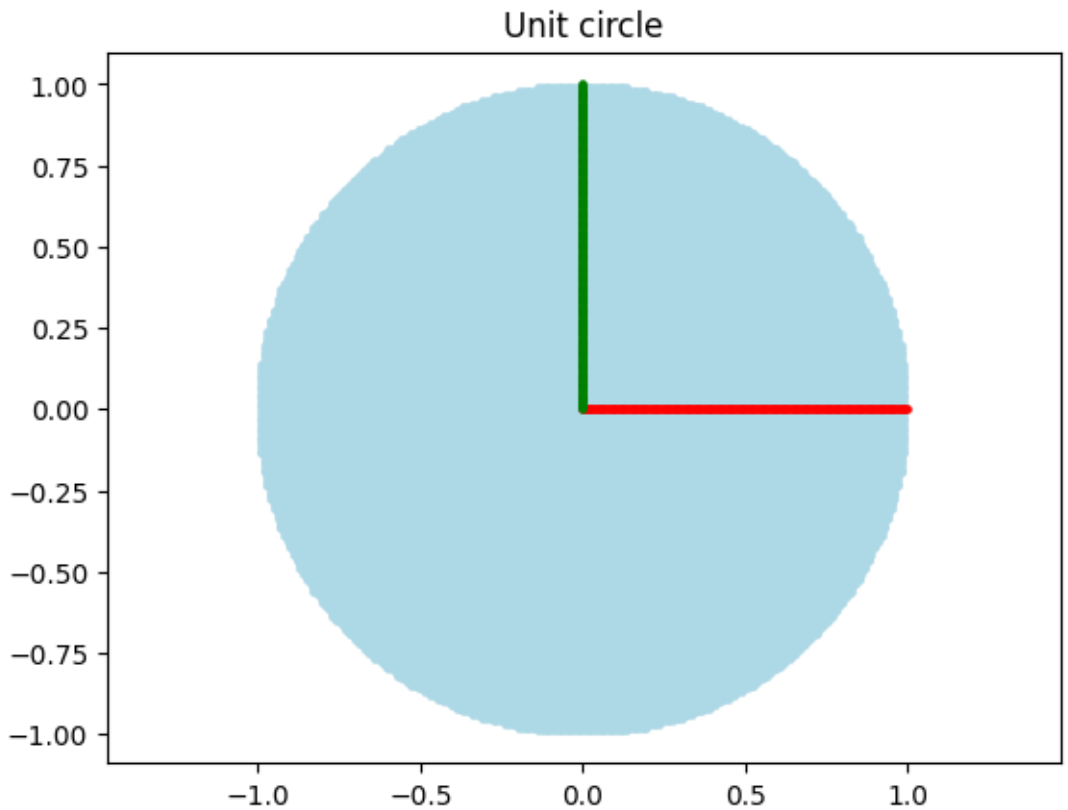
2.3 V^T as a composition of reflection and rotation matrix

In general an orthonormal transformation can be viewed as compositions of rotation and reflection operators. Next we observe the effect of setting

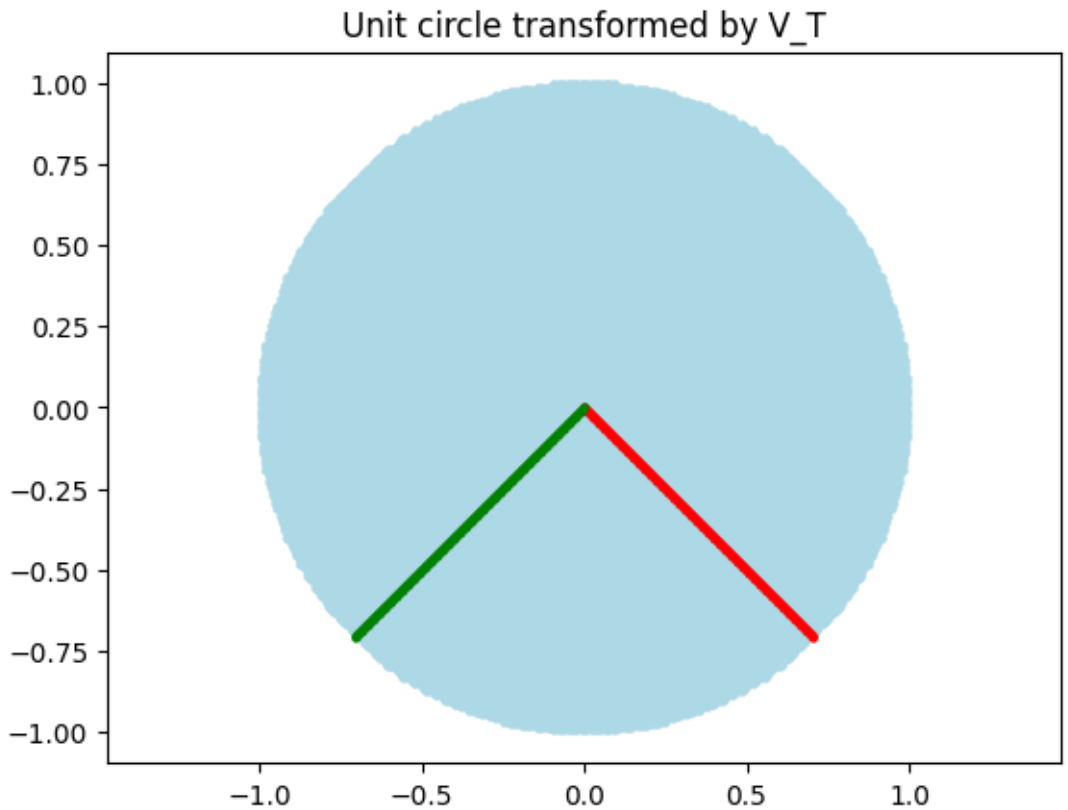
$V^T = RFx() \cdot RCC\left(\frac{\pi}{4}\right)$

[26]:
```
VT_4 = VT_3 @ VT_1
#Check that VT_4 is still orthonormal
print("VT_4 is orthonormal?: ", is_orthonormal(VT_4))
visualize(VT = VT_4, show_DVT=False, show_UDVT=False)
```

```
VT_4 is orthonormal?:  True
```

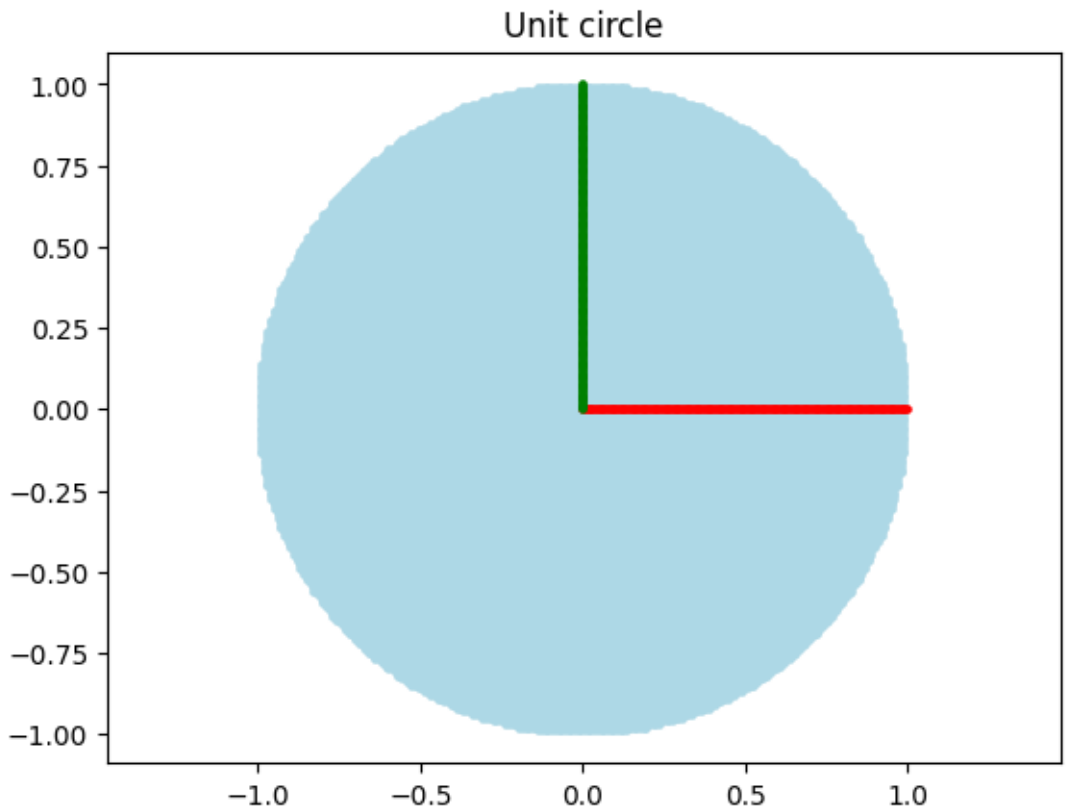



```
WARNING:matplotlib.axes._base:Ignoring fixed x limits to fulfill fixed data
aspect with adjustable data limits.
```

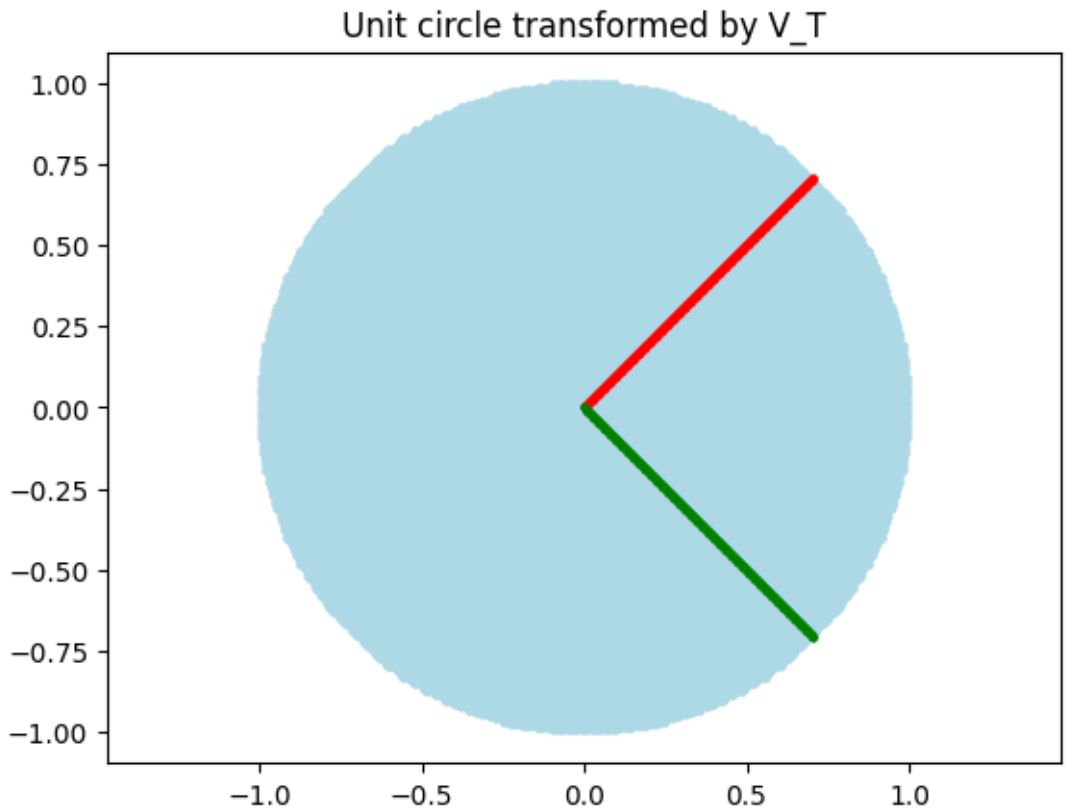


2.3.1 (b) iii: Comment on the effect of $V^T = RCC\left(\frac{\pi}{4}\right) \cdot RFx()$. Is it same as the case when $V^T = RFx() \cdot RCC\left(\frac{\pi}{4}\right)$?

[27]:
```
VT_5 = VT_1 @ VT_3
visualize(VT = VT_5, show_DVT=False, show_UDVT=False)
```




```
WARNING:matplotlib.axes._base:Ignoring fixed x limits to fulfill fixed data
aspect with adjustable data limits.
```



2.3.2 #TODO: Fill in solution to (b) iii here

No the effect is different, the final vector lands on different coordinates. One is rotation then reflection, the other is reflection then rotation. Order matters.

3 Effect of linear transformation by diagonal matrix D

The diagonal matrix D with entries σ_1 and σ_2, transforms the unit circle into an ellipse with x direction scaled by σ_1 and y direction scaled by σ_2.

If $\sigma_1 > \sigma_2$, then the major axis of the ellipse will be along the x-axis.

If $\sigma_1 < \sigma_2$, then the major axis of the ellipse will be along the y-axis.

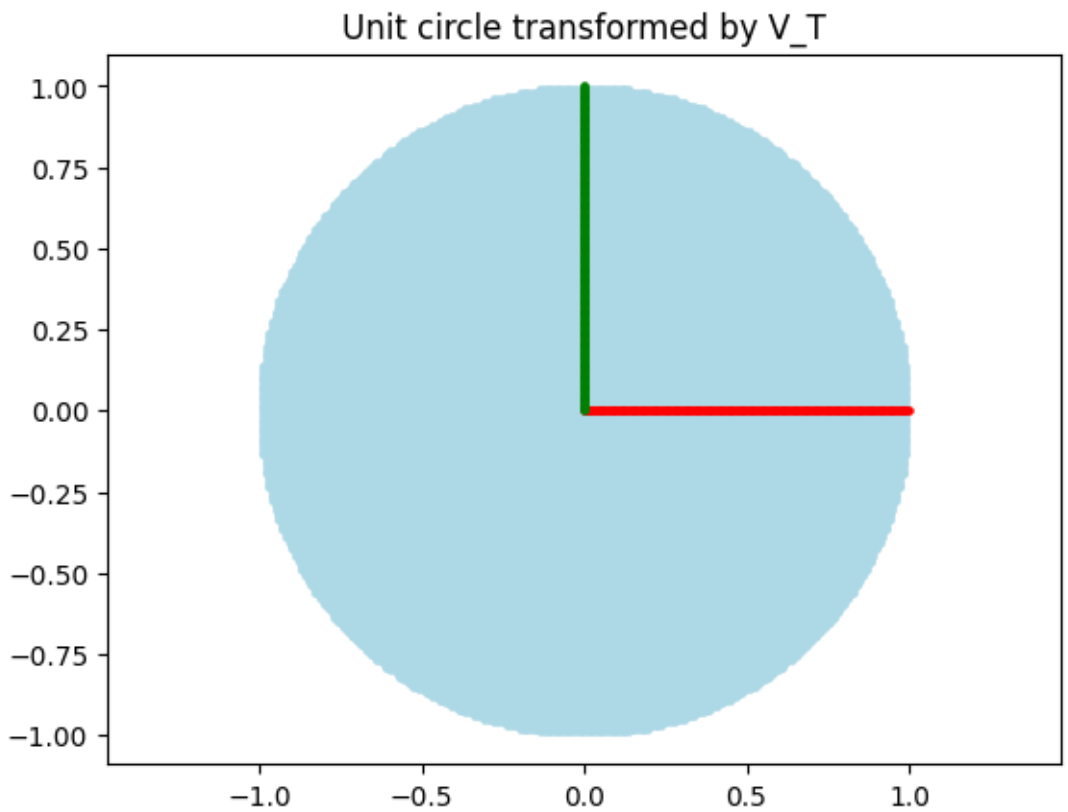
If $\sigma_1 = \sigma_2$, then the ellipse will have both axis equal (i.e it is a circle).

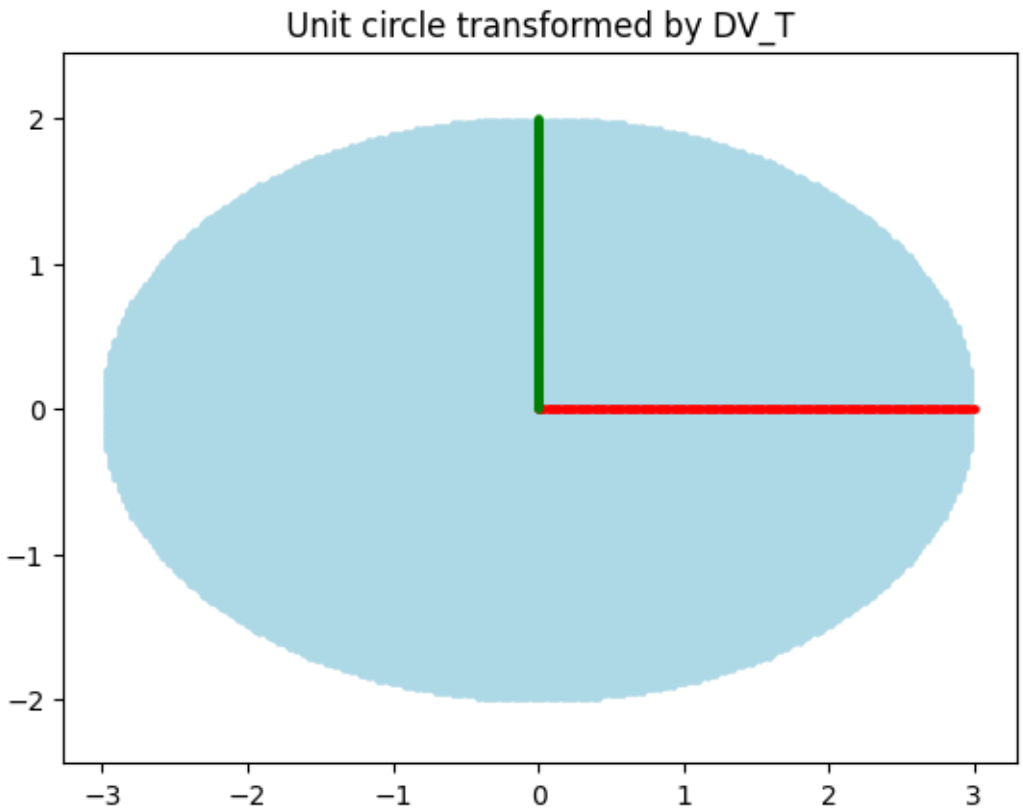
Note that multiplying by D, does not rotate or reflect points in any way. It is a purely scaling operation where different directions get scaled by different values based on entries of D.

3.0.1 (c) i: Comment on the length of major and minor axis of the ellipse and their orientation with respect to X and Y axis when D has entries [3, 2]. Here V is identity.

[28]:
```
D_1 = np.array([3, 2])
visualize( D = D_1, show_original=False, show_UDVT=False)
```

```
WARNING:matplotlib.axes._base:Ignoring fixed x limits to fulfill fixed data
aspect with adjustable data limits.
```





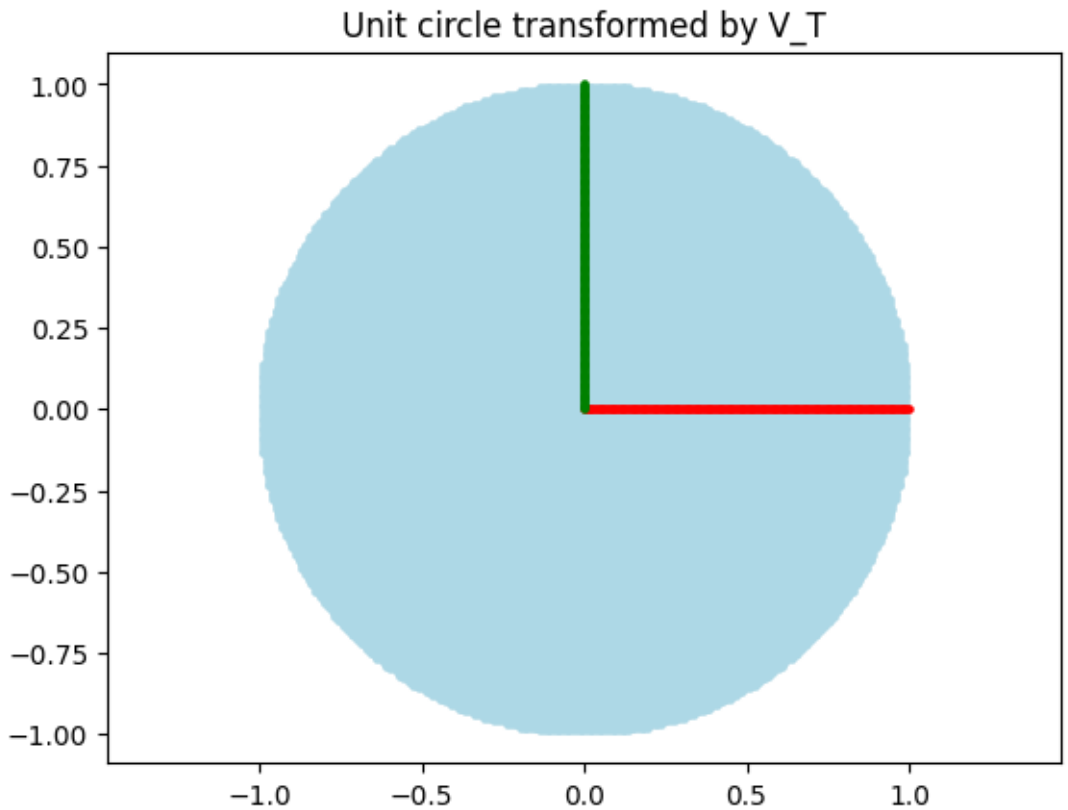
3.0.2 #TODO: Fill in solution to (c) i here

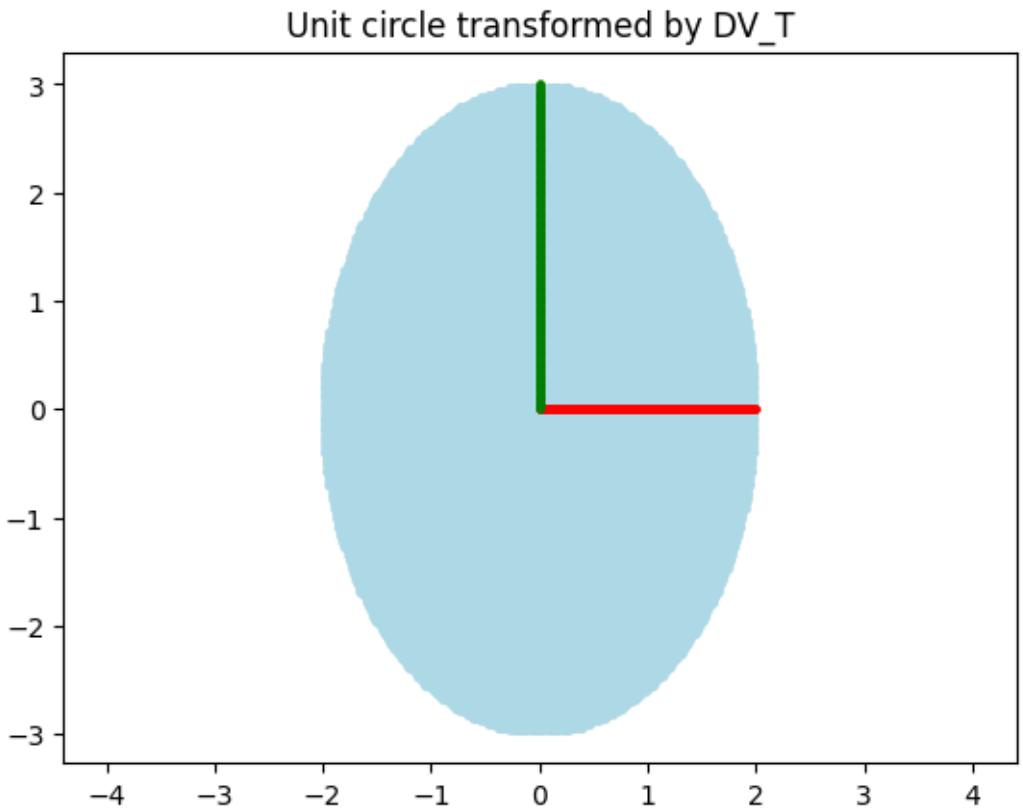
Since $\sigma_1 > \sigma_2$ $(3 > 2)$, the major axis of the ellipse will be along the x-axis. Length along x-axis is longer compared to y-axis.

3.0.3 (c) ii: Comment on the length of major and minor axis of the ellipse and their orientation with respect to X and Y axis when D has entries [2, 3]. Here V is the identity matrix.

[29]:
```
D_2 = np.array([2, 3])
visualize( D = D_2,  show_original=False, show_UDVT=False)
```

```
WARNING:matplotlib.axes._base:Ignoring fixed x limits to fulfill fixed data
aspect with adjustable data limits.
```


Unit circle transformed by V_T
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.0
−0.5
0.0
0.5
1.0



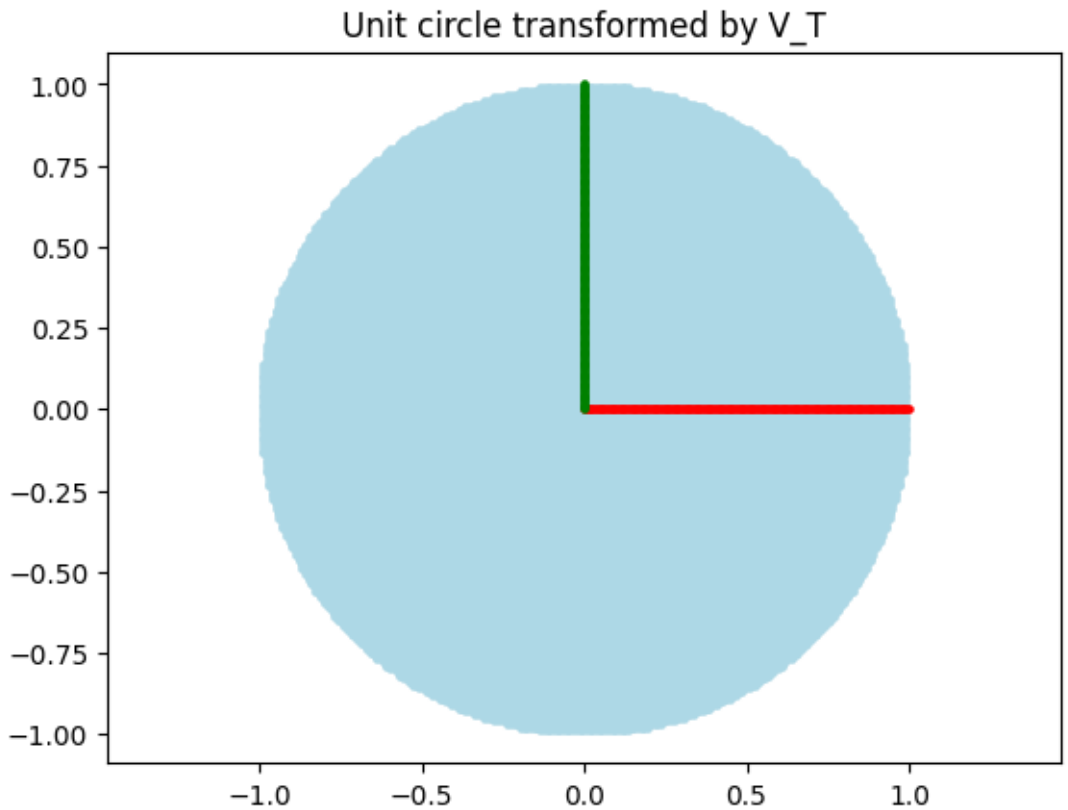
3.0.4 #TODO Fill in solution to (c) ii here

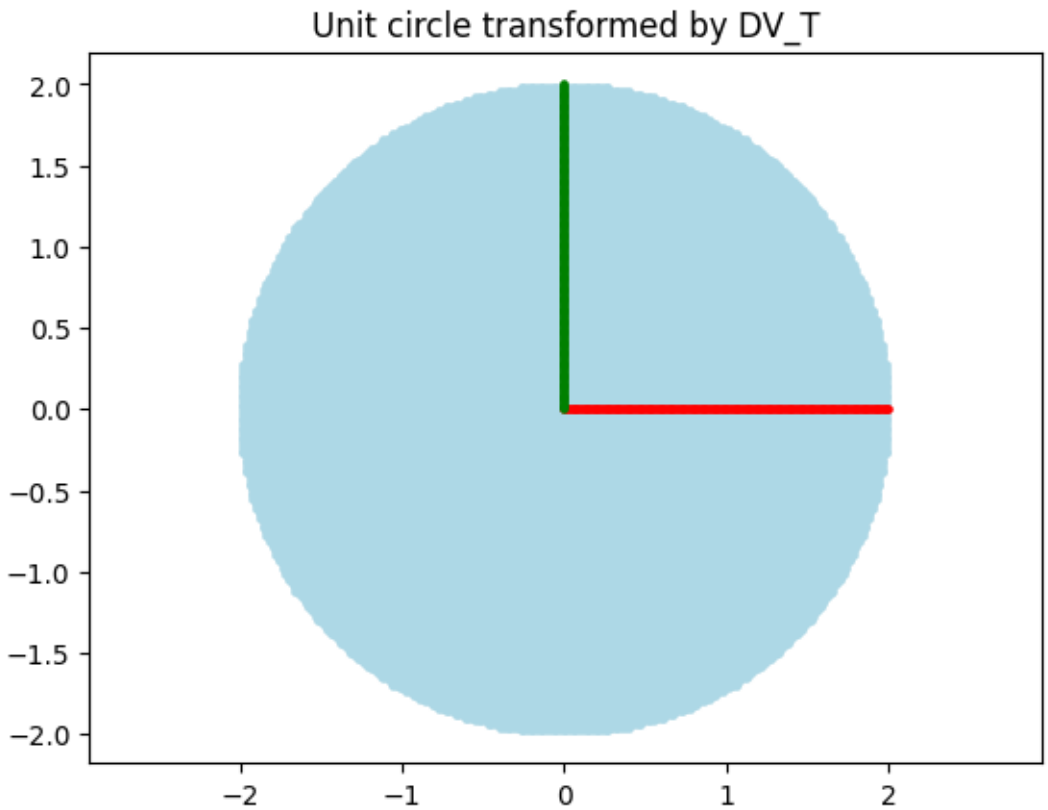
Since 2<3 ($\sigma 1 < \sigma 2$), the major axis of the ellipse is along the y-axis.

3.0.5 (c) iii: What can you say about the ellipse when D has entries [2, 2]? Here V is the identity matrix.

[30]:
```
D_3 = np.array([2, 2])
visualize( D = D_3, show_original=False, show_UDVT=False)
```

```
WARNING:matplotlib.axes._base:Ignoring fixed x limits to fulfill fixed data
aspect with adjustable data limits.
```


Unit circle transformed by V_T
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.0
−0.5
0.0
0.5
1.0



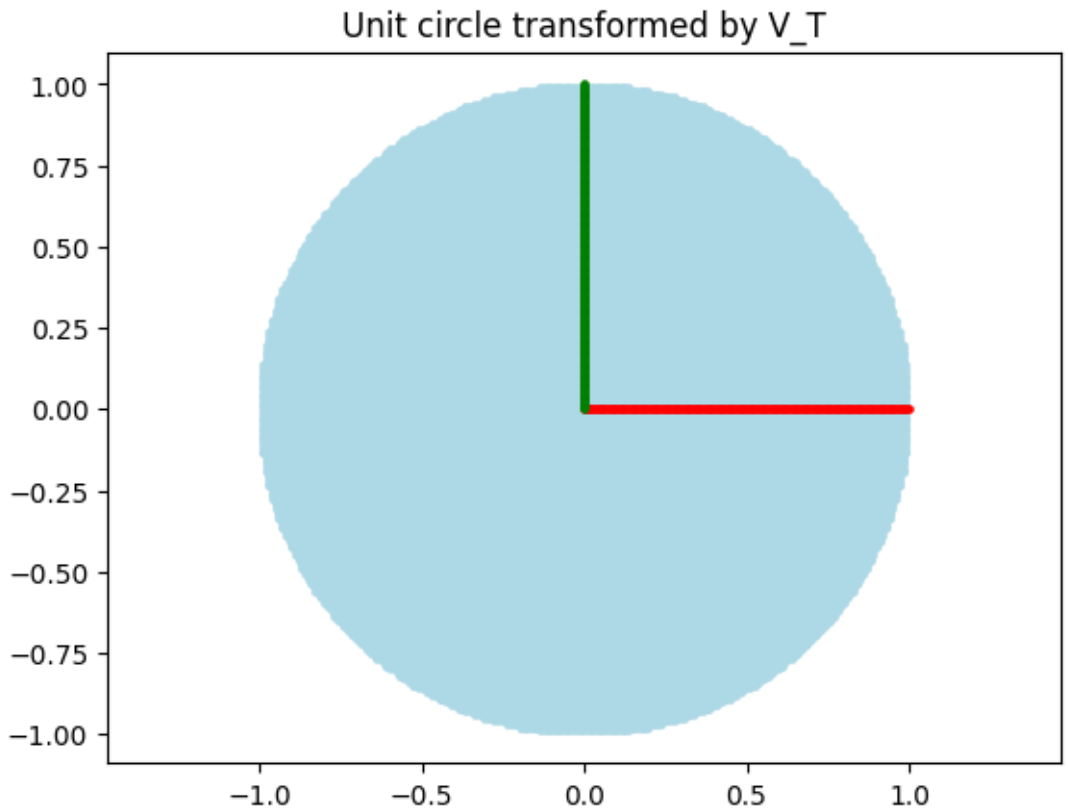
3.0.6 #TODO Fill in solution to (c) iii here

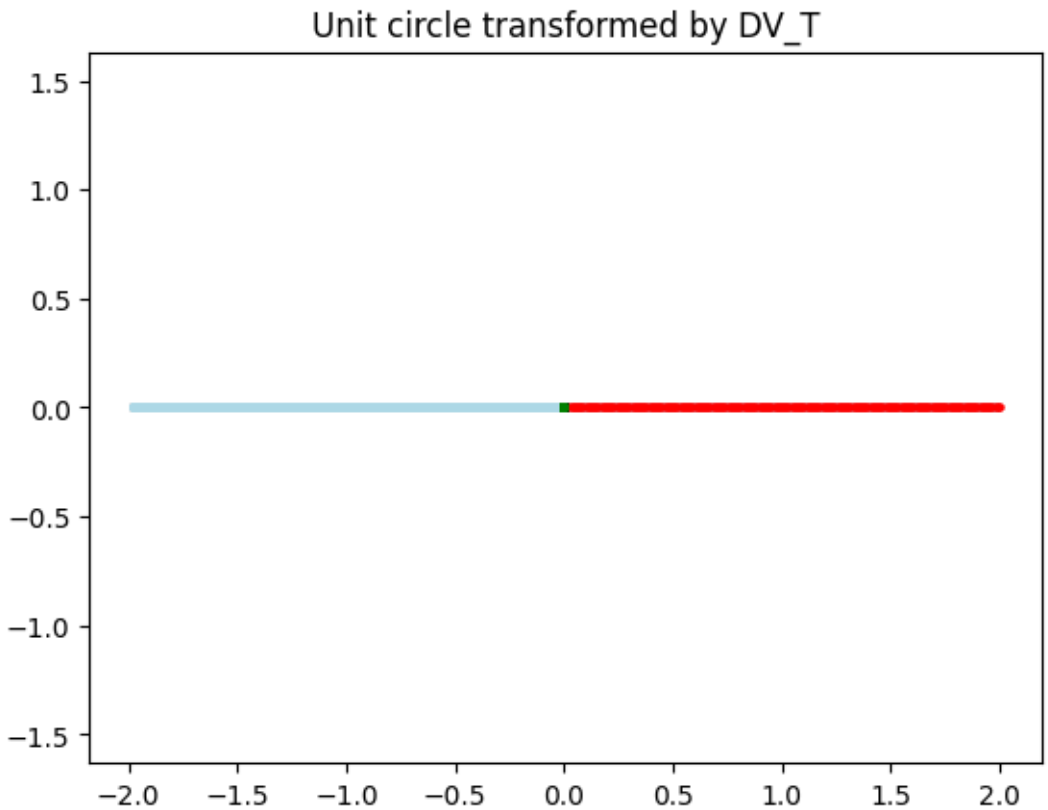
Since $\sigma 1 = \sigma 2$ $(2 = 2)$, the ellipse have both axis equal (i.e it is a circle).

3.0.7 (c) iv: What can you say about the ellipse when D has entries [2, 0]? Here V is the identity matrix.

[31]:
```
D_4 = np.array([2, 0])
visualize( D = D_4, show_original=False, show_UDVT=False)
```

```
WARNING:matplotlib.axes._base:Ignoring fixed x limits to fulfill fixed data
aspect with adjustable data limits.
```


Unit circle transformed by V_T
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.0
−0.5
0.0
0.5
1.0



3.0.8 #TODO: Fill in solution to (c) iv here

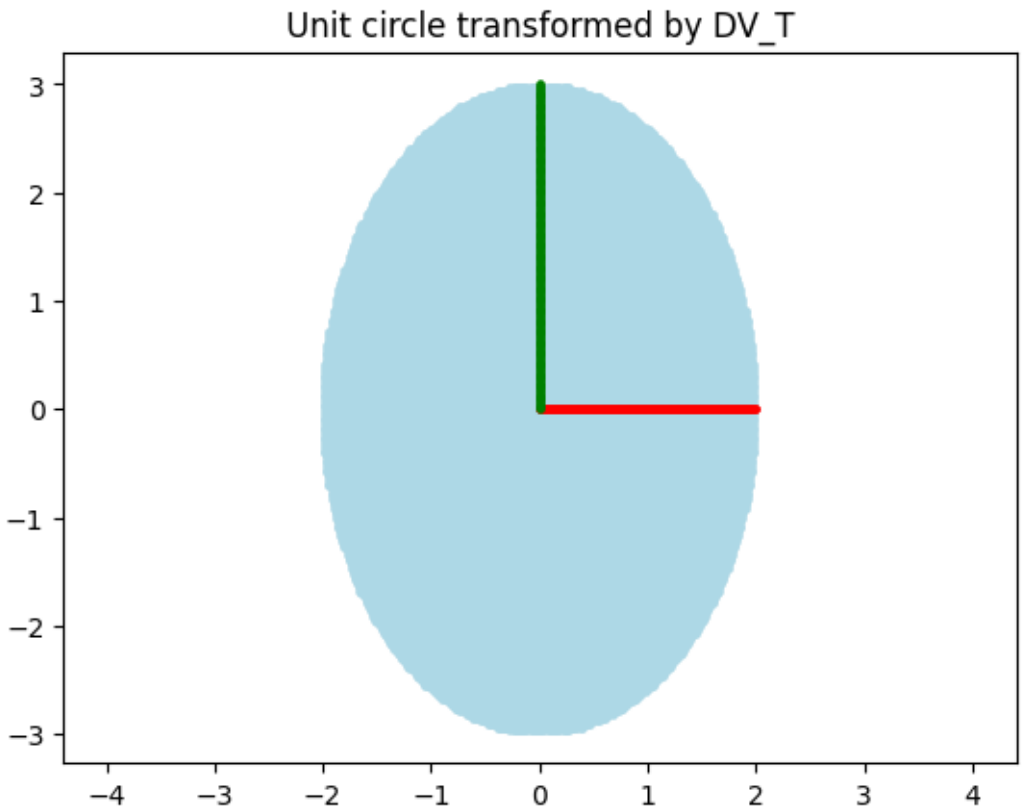
Since the y component is scaled by a factor of 0, and the x component is scaled by 2, it becomes a straight line with major axis in the x-axis.

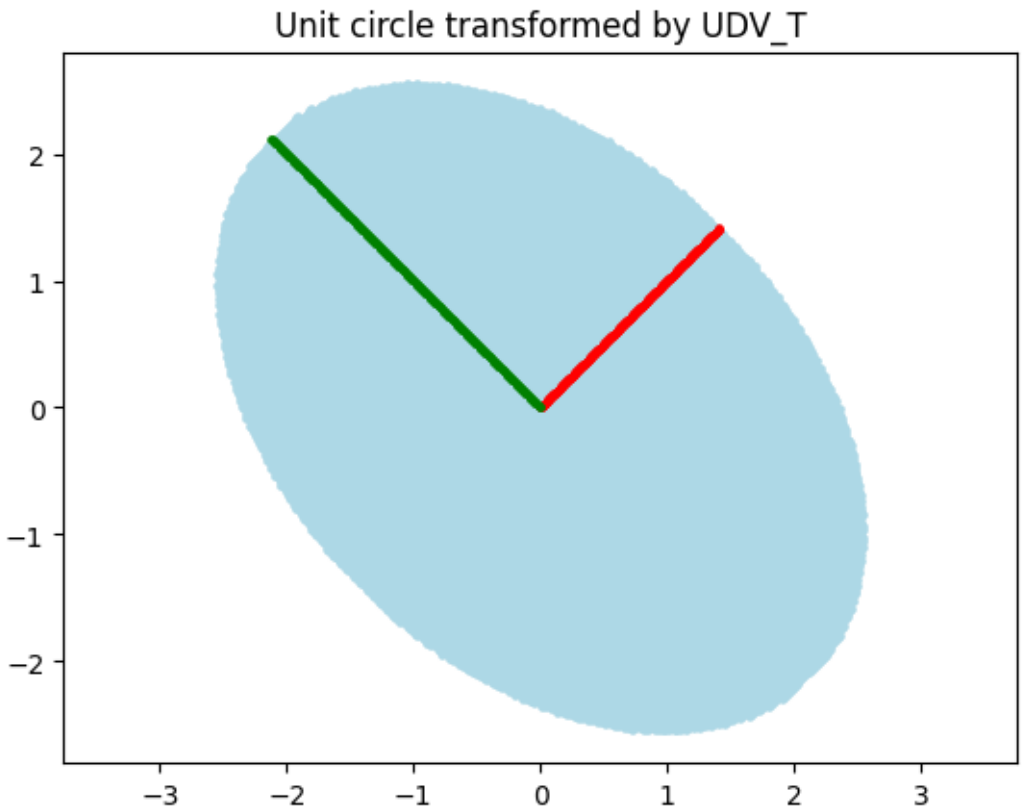
4 Effect of the linear transformation by an orthonormal matrix U

As we saw before for V^T, a 2 x 2 orthonormal matrix can be viewed as a linear transformation that performs some combination of rotations and reflections.

4.0.1 (d) i: Comment on the effect of $U = RCC\left(\frac{\pi}{4}\right)$ as in cell below. The value of D is in the code below and V is the identity matrix. What happened to the ellipse? Did the length of the major and minor axis change?

[32]:
```
U_1 = get_RCC(np.pi/4)
visualize( U = U_1, D =np.array([2,3]), show_original=False, show_VT=False)
```


Unit circle transformed by DV_T
3
2
1
0
−1
−2
−3
−4
−3
−2
−1
0
1
2
3
4

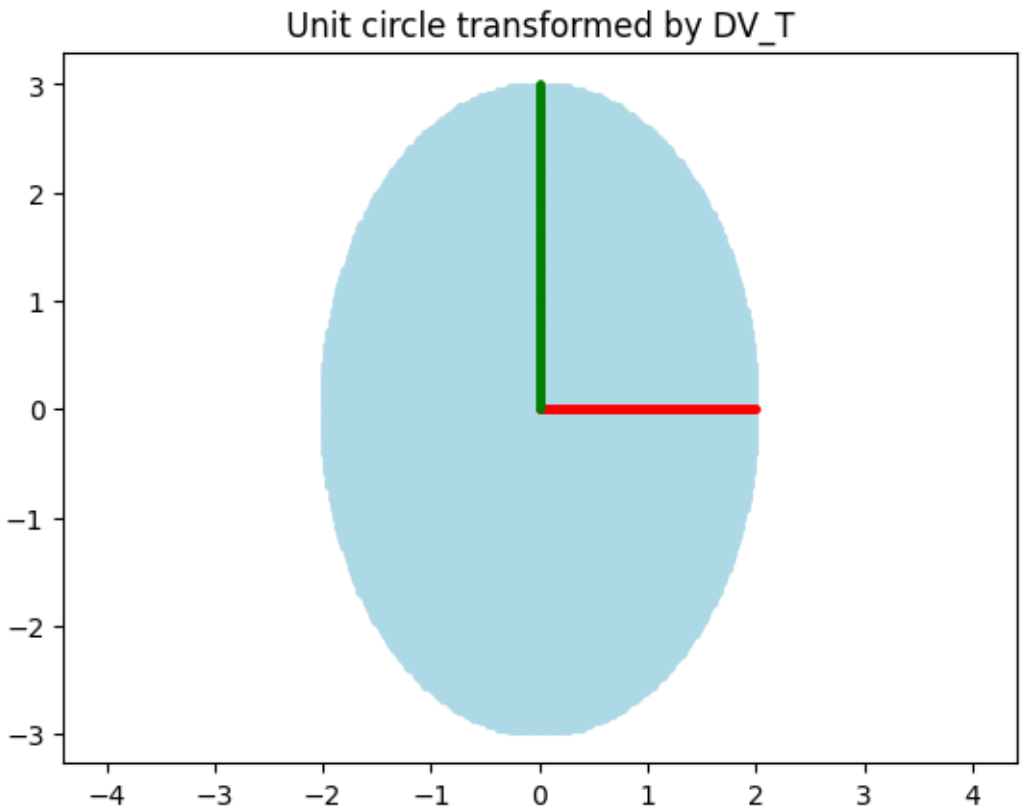


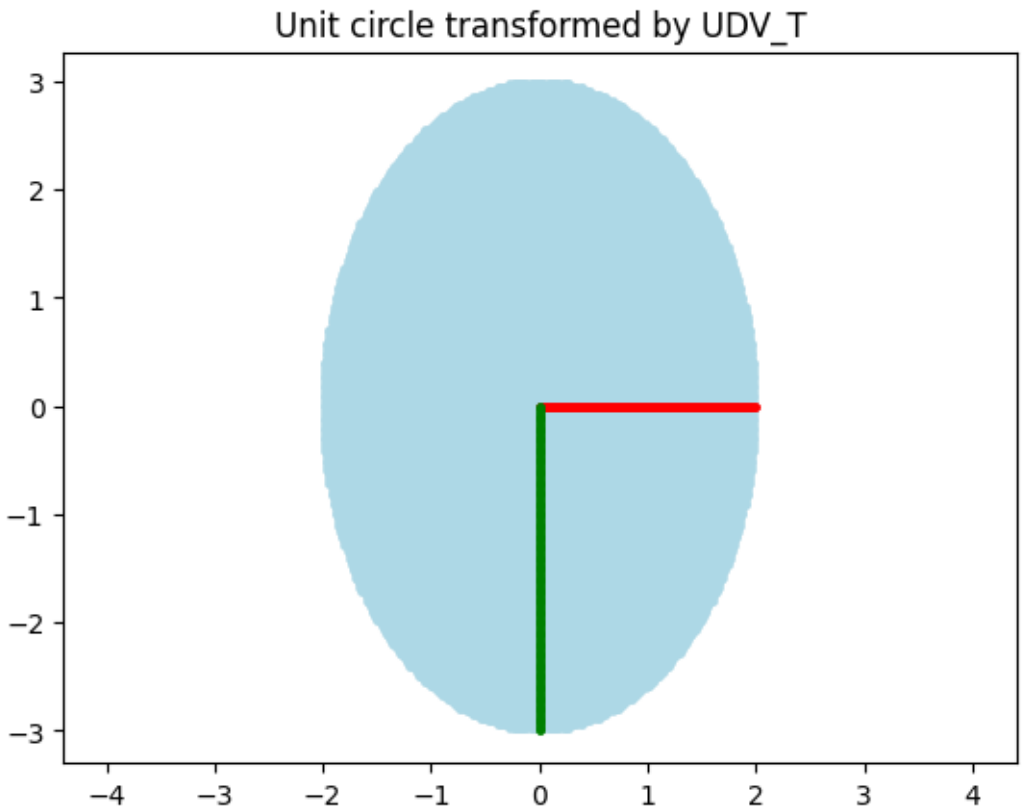
4.0.2 #TODO: Fill in solution to (d) i here

The ellipse is rotated by set theta degrees. The length of the major and minor axis does not change.

4.0.3 (d) ii: Comment on the effect of $U = RFx()$ as in cell below. The value of D is in the code below and V is the identity matrix. What happened to the ellipse? Did length of major and minor axis change?

[33]:
```
U_2 = get_RFx()
visualize( U = U_2, D =np.array([2,3]), show_original=False, show_VT=False)
```


Unit circle transformed by DV_T
3
2
1
0
−1
−2
−3
−4
−3
−2
−1
0
1
2
3
4



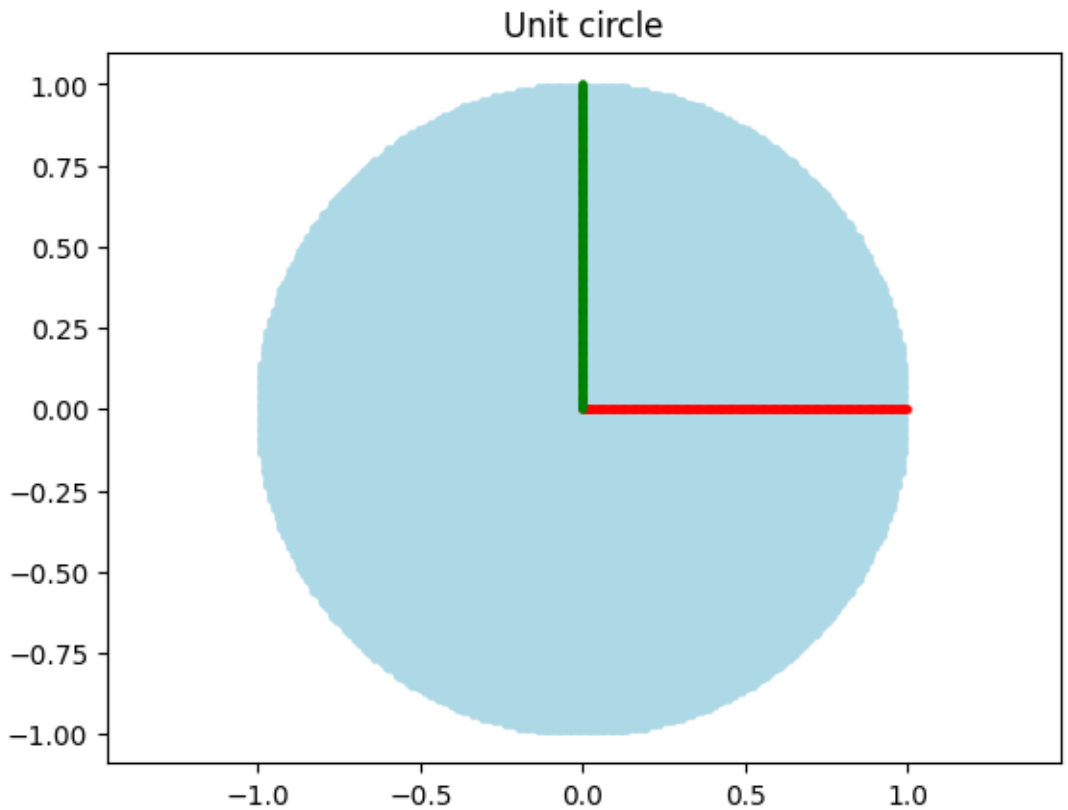
4.0.4 #TODO Fill in solution to (d) ii here

Ellipse rotated along the x-axis, length of both axis stayed the same

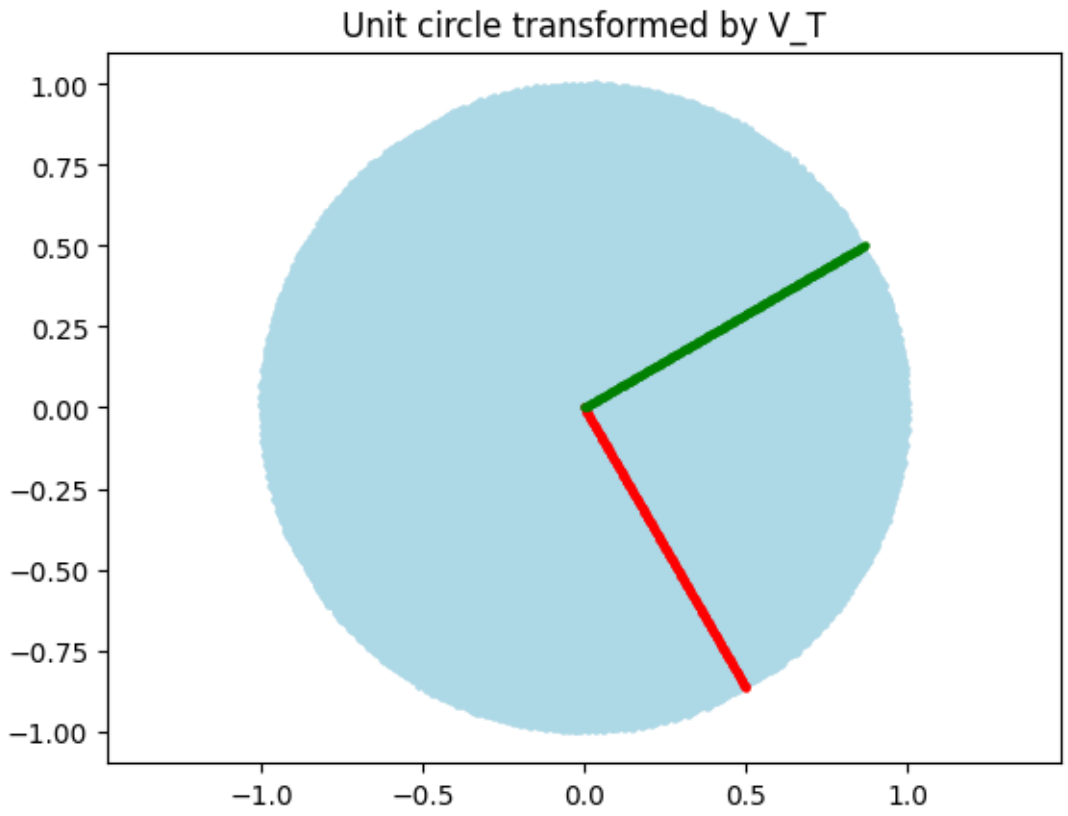
5 Putting everything together. Effect of linear transformation by UDV^T

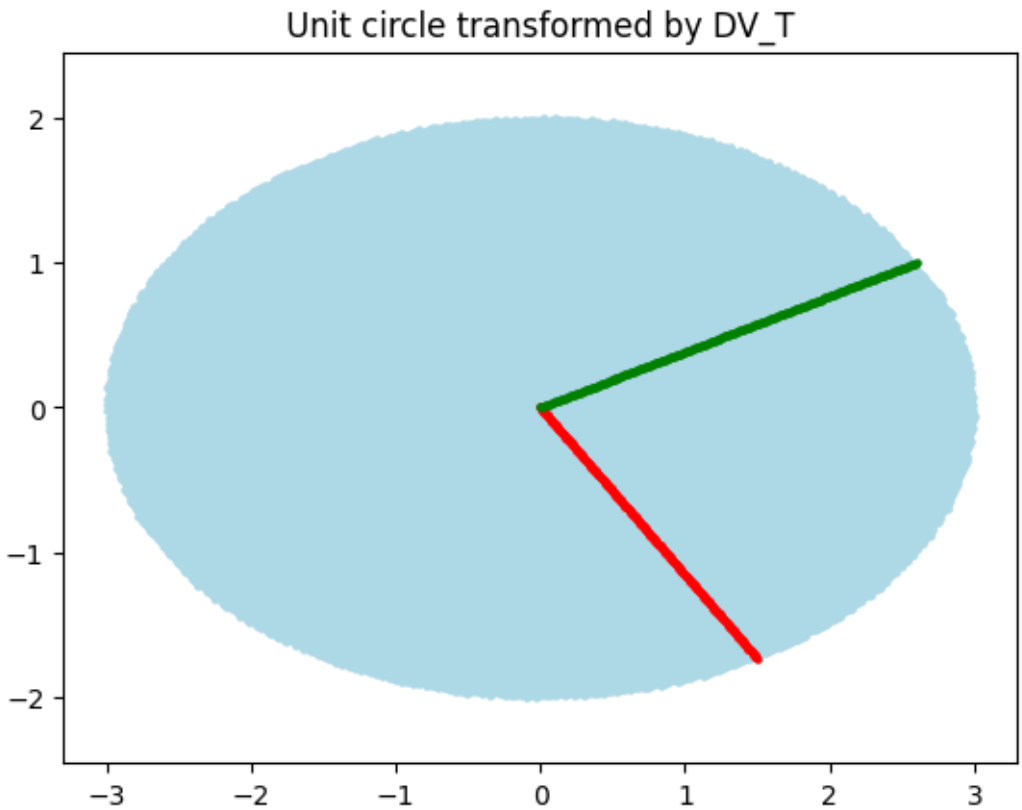
5.0.1 Case I

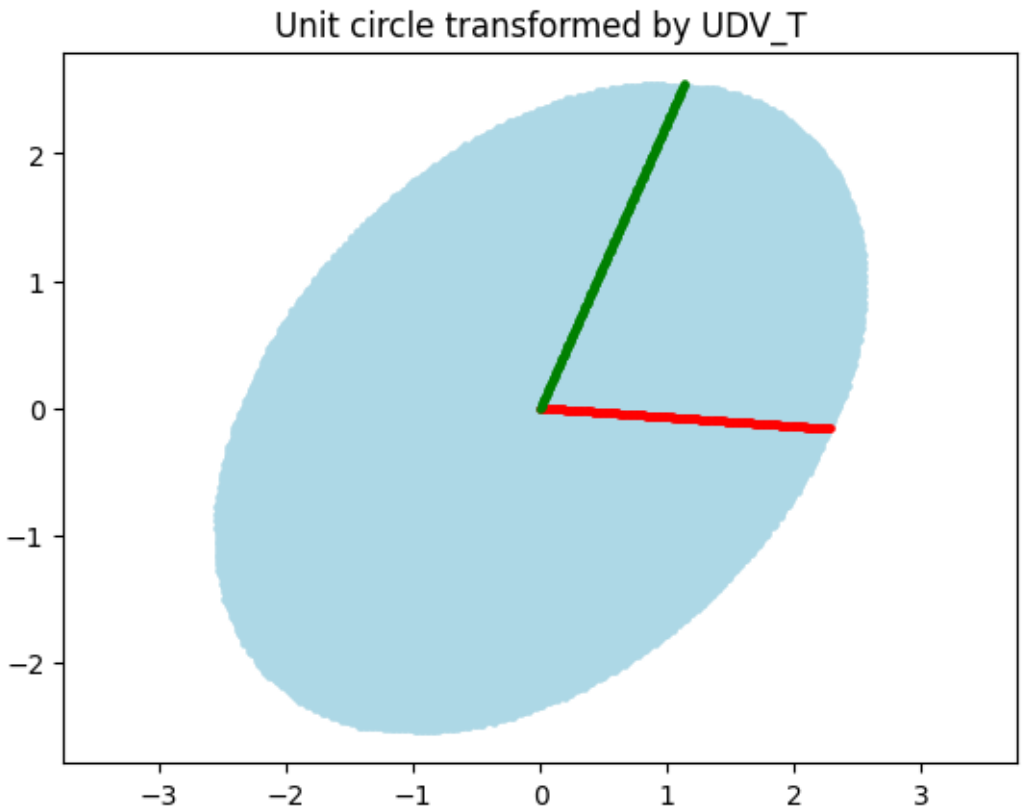
[34]:
```
U = get_RCC(np.pi/4)
VT = get_RCC(-np.pi/3)
D = np.array([3,2])
visualize(U = U, VT= VT, D=D)
```




```
WARNING:matplotlib.axes._base:Ignoring fixed x limits to fulfill fixed data
aspect with adjustable data limits.
```


Unit circle transformed by V_T
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.0
−0.5
0.0
0.5
1.0

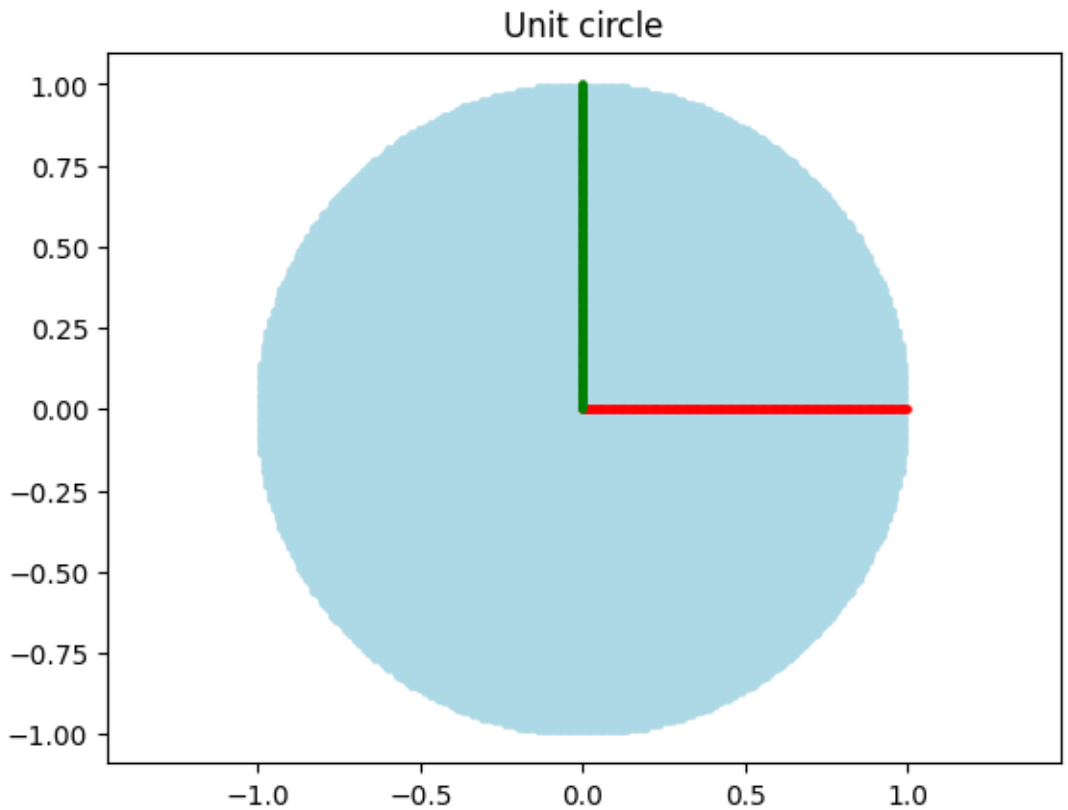

Unit circle transformed by DV_T
2
1
0
−1
−2
−3
−2
−1
0
1
2
3



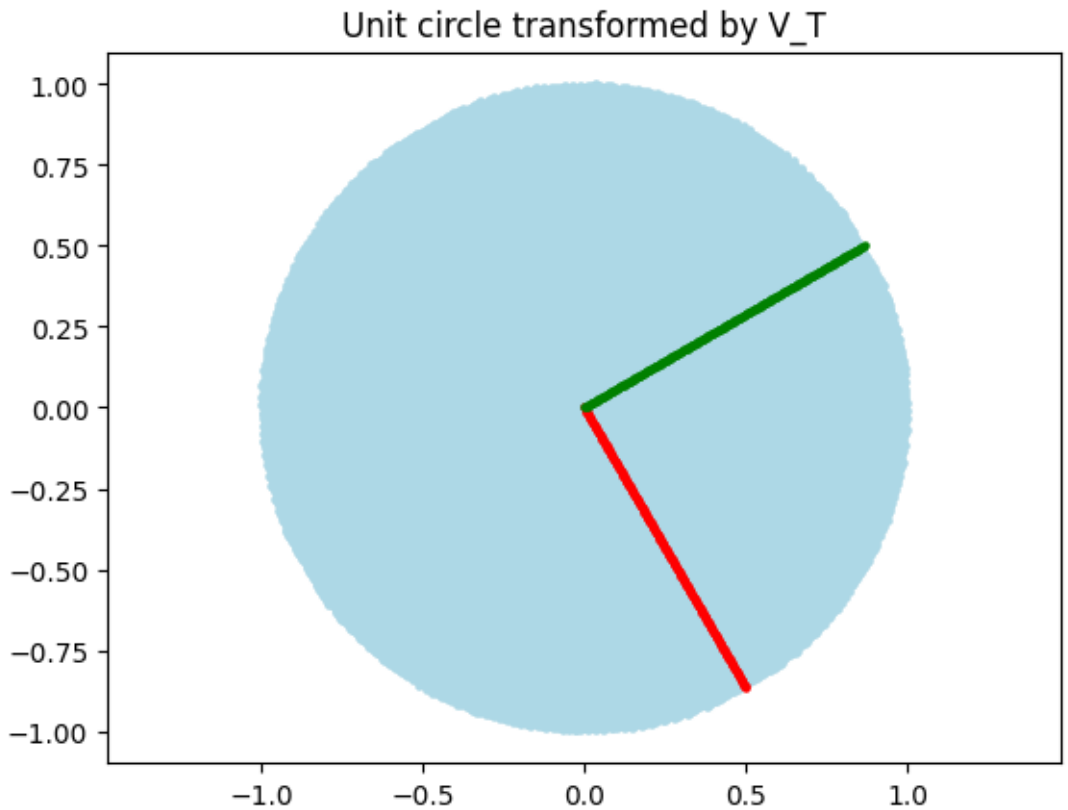
The above figures show the transformation after each step.

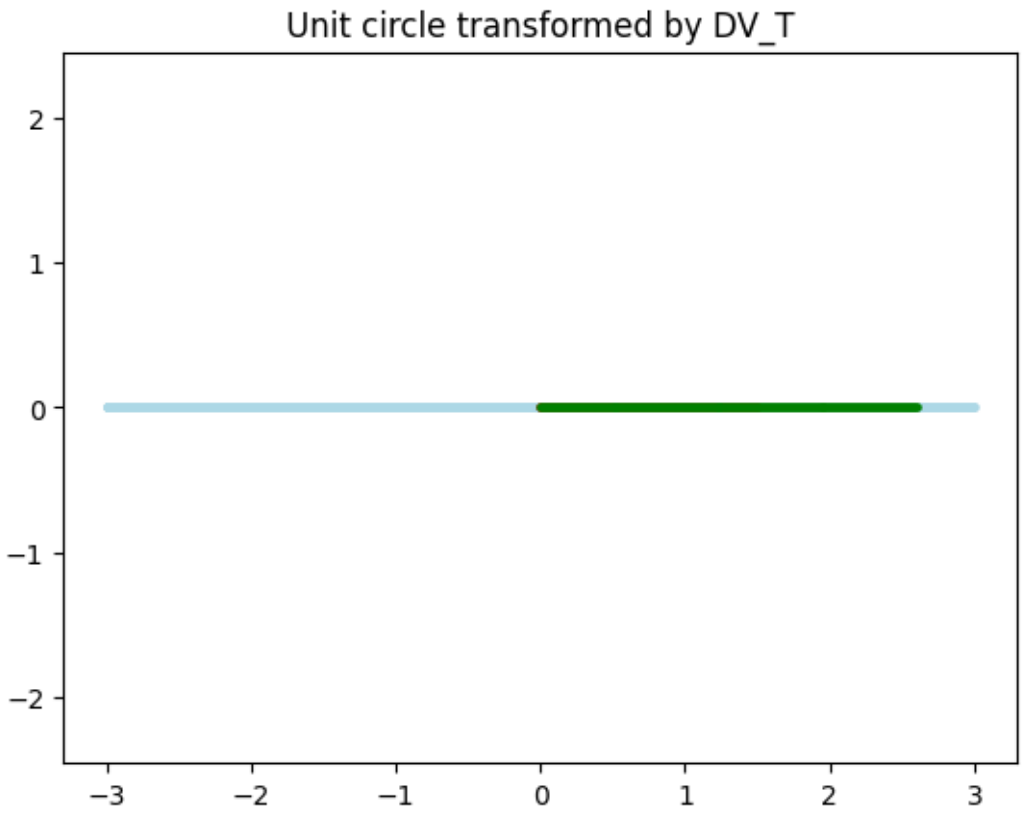
5.0.2 Case II

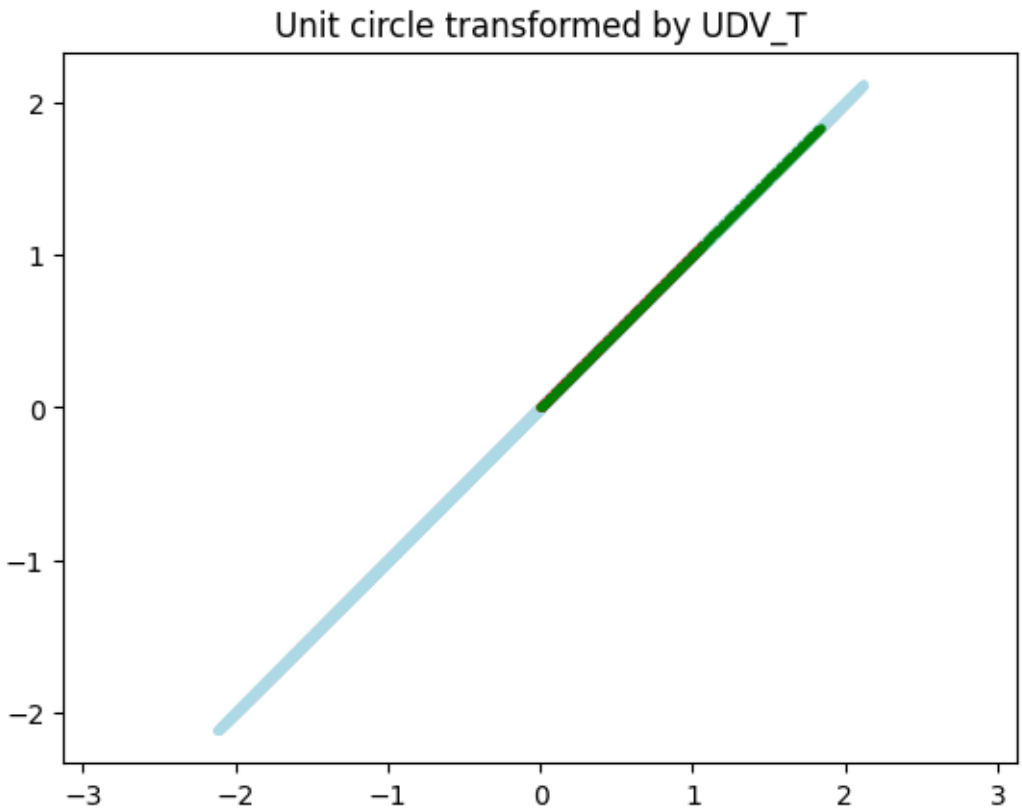
[35]:
```
U = get_RCC(np.pi/4)
VT = get_RCC(-np.pi/3)
D = np.array([3,0])
visualize(U = U, VT= VT, D=D)
```




```
WARNING:matplotlib.axes._base:Ignoring fixed x limits to fulfill fixed data
aspect with adjustable data limits.
```


Unit circle transformed by V_T
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.0
−0.5
0.0
0.5
1.0

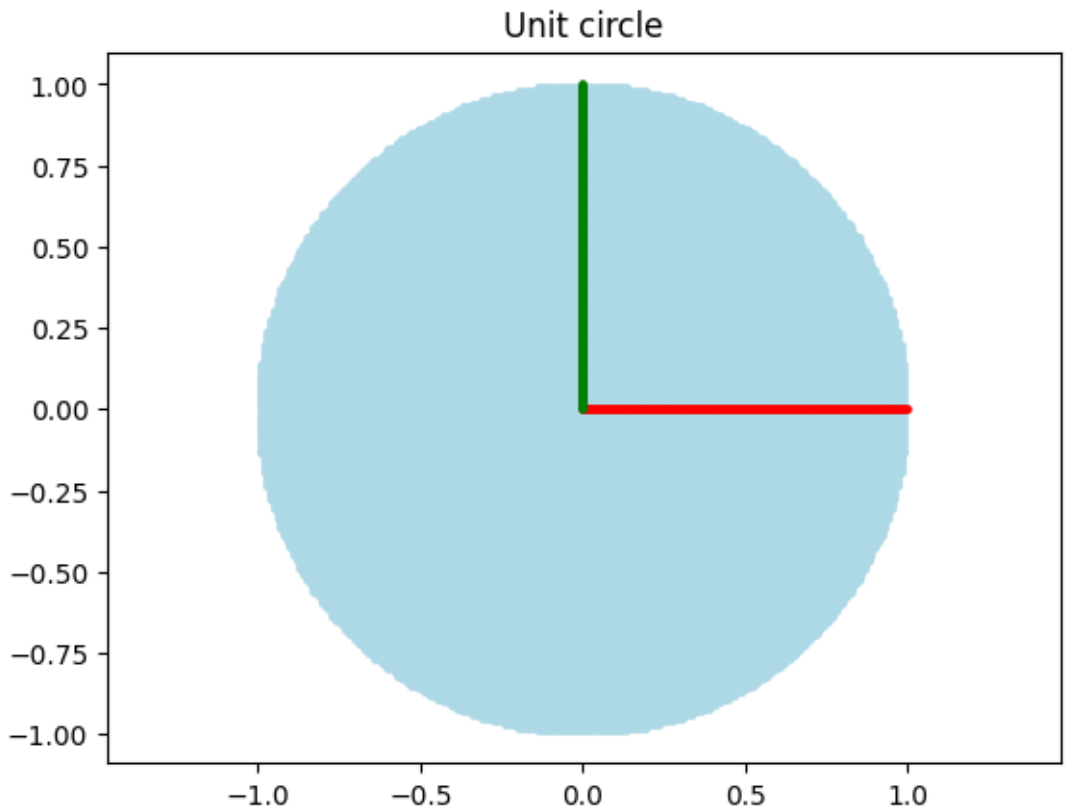

Unit circle transformed by DV_T
2
1
0
−1
−2
−3
−2
−1
0
1
2
3



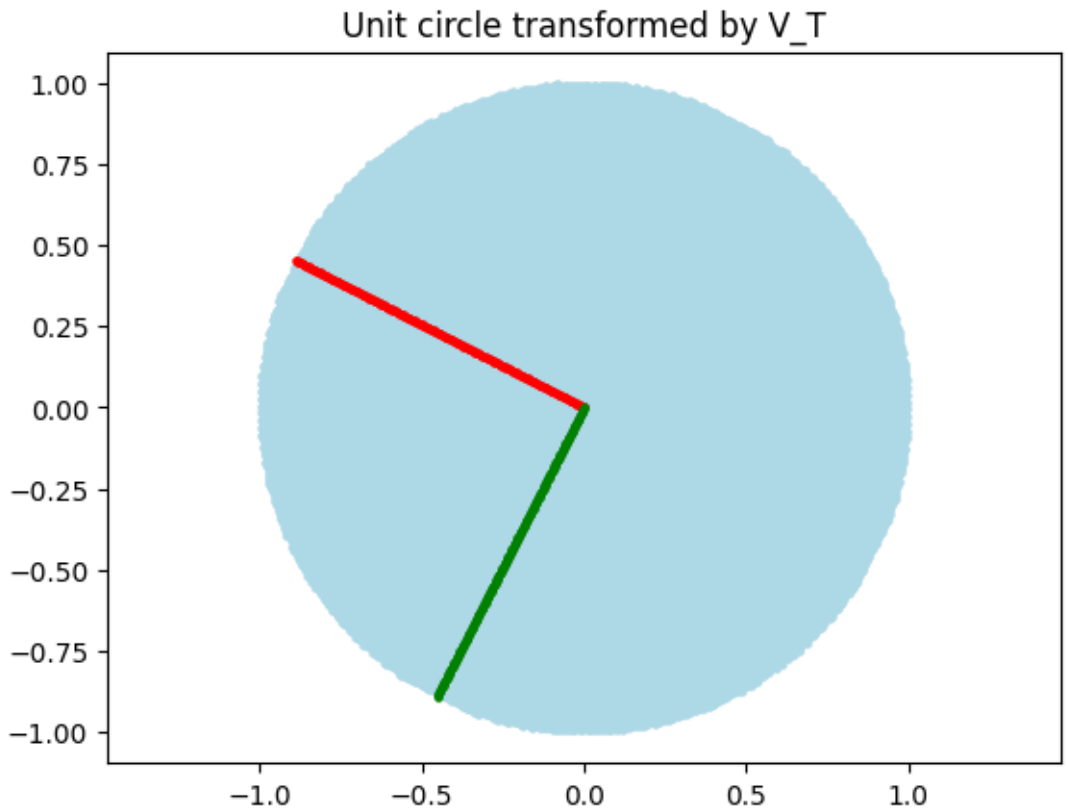
The above figures show the transformation after each step.

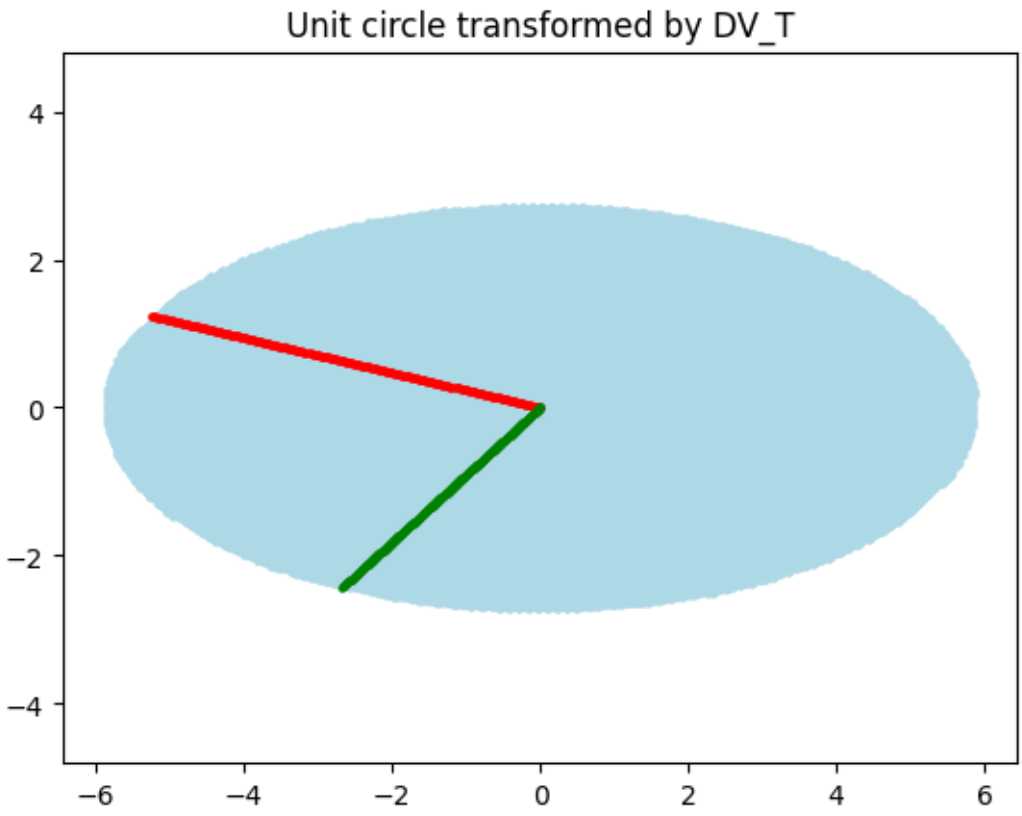
5.0.3 Case III

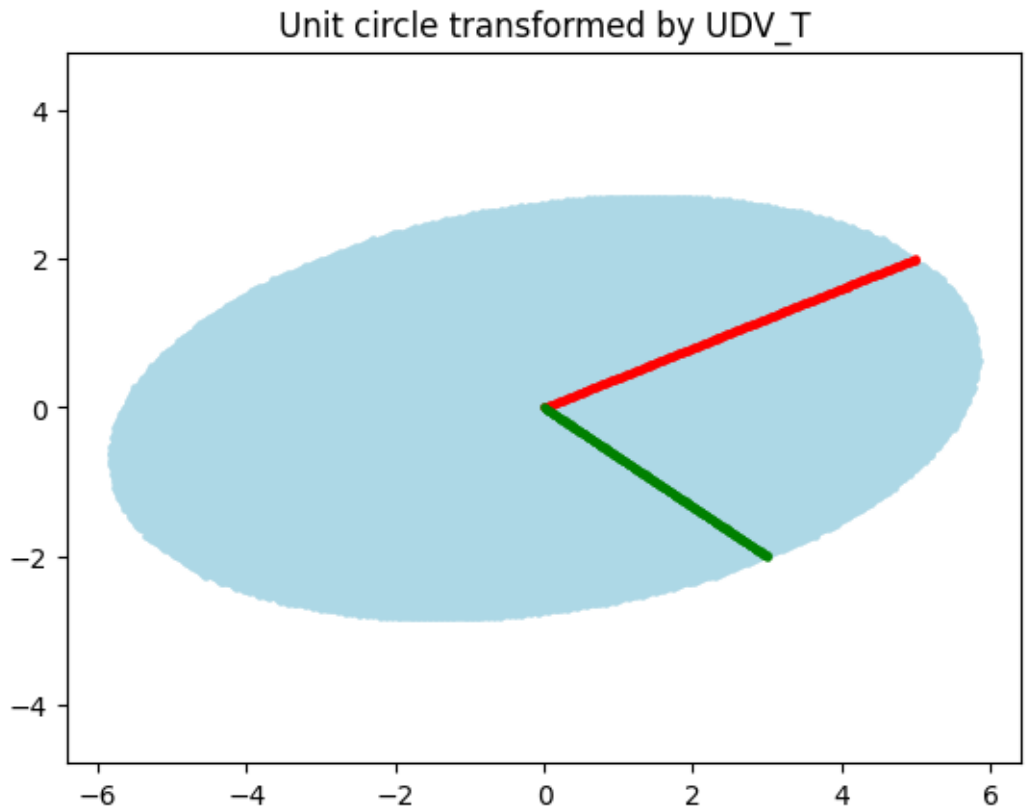
[36]:
```
A = np.array([[5, 3], [2, -2]])
U,D,VT = np.linalg.svd(A)
visualize(U = U, D=D, VT=VT)
```




```
WARNING:matplotlib.axes._base:Ignoring fixed x limits to fulfill fixed data
aspect with adjustable data limits.
```


Unit circle transformed by V_T
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.0
−0.5
0.0
0.5
1.0


Unit circle transformed by DV_T
4
2
0
−2
−4
−6
−4
−2
0
2
4
6



5.0.4 (e): For case III, based on the figures obtained by running the cell, answer the following questions:

1) Is V^T a pure rotation, pure reflection or combination of both?
2) Let σ_1 and σ_2 denote the entries of the diagonal matrix in SVD of A, with $\sigma_1 > \sigma_2$? What is an approximate value of $\frac{\sigma_1}{\sigma_2}$?
3) Is U a pure rotation, pure reflection or combination of both?

5.0.5 #TODO Fill in solution to (e) here

1) pure rotation counterclockwise of about 150 deg
2) length of major (x) axis 12, minor (y) axis 6, 12/6 = 2
3) combination of both, first reflection across x-axis, then counterclockwise rotation of 180 deg

[47]:
```
%%capture
from google.colab import drive
drive.mount('/content/drive')
!apt-get install texlive-xetex texlive-fonts-recommended texlive-plain-generic␣
↪-qq
```

```
!pip install nbconvert
import os
os.chdir("/content/drive/MyDrive/Colab Notebooks/")
!jupyter nbconvert --to pdf "hw4_svd_transformation.ipynb"
```

6 Exploration Area (Not part of homework question)

You are free to visualize the effect of the SVD transformation on the unit circle for whatever matrix you desire

[ ]:
```
# #Sample format 1
# U = get_RCC(np.pi/4)
# VT = get_RCC(-np.pi/3)
# D = np.array([3,2])
# visualize(U = U, VT= VT, D=D)
```

[ ]:
```
# #Sample format 2
# A = np.array([[5, 3], [2, -2]])
# U,D,VT = np.linalg.svd(A)
# visualize(U = U, D=D, VT=VT)
```